



SAINT IGNATIUS' COLLEGE

Trial Higher School Certificate

2007

MATHEMATICS EXTENSION 1

8:45am – 10:50am
Friday 31st August 2007

Directions to Students

• Reading Time: 5 minutes	• Total Marks: 84
• Working Time: 2 hours	
• Write using blue or black pen (sketches in pencil).	• Attempt Questions 1 – 7
• Board approved calculators may be used	• All questions are of equal value
• A table of standard integrals is provided at the back of this paper.	
• All necessary working should be shown in every question.	
• Answer each question in the booklets provided and clearly label your name and teacher's name.	

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QUESTION 1 (use a SEPARATE writing booklet)

- (a) Given that if $y = \sin^{-1}\left(\frac{x}{a}\right)$ then $\frac{dy}{dx} = \frac{1}{\sqrt{a^2 - x^2}}$, where $x < |a|$,

write down an expression for $\frac{dy}{dx}$ if y equals:

- (i) $\sin^{-1}(x)$
- (ii) $\sin^{-1}\left(\frac{x}{7}\right)$
- (iii) $\sin^{-1}(7x)$ (3M)

- (b)
- (i) If $f(x) = x^3 - 3x^2 - 4x + 12$, show that $f(3) = 0$ (1M)
- (ii) Hence solve $x^3 - 3x^2 - 4x + 12 = 0$ (2M)

- (c) Prove that
- $$\frac{1}{3}[n(n+1)(n+2)] + (n+1)(n+2) = \frac{1}{3}(n+1)(n+2)(n+3)$$
- (2M)

- (d) Solve for x
- $$\frac{x-2}{x+4} \geq \frac{1}{3}$$
- (4M)

QUESTION 2 (use a SEPARATE writing booklet)

- (a) A particle P moves in a straight line so that its distance x from a central fixed point O at time t is given by

$$x = 2 \sin \left(5t + \frac{\pi}{6} \right)$$

- (i) Write down an expression for the velocity $\dot{x}$
- (ii) Write down an expression for the acceleration $\ddot{x}$
- (iii) The particle P is said to be executing Simple Harmonic Motion. Explain why this is so.
- (3M)

- (b) The parametric equations of a parabola are

$$x = 2t$$

$$y = 2t^2$$

Find the equation of the tangent to this parabola at the point

$$P(2, 2) \quad (2M)$$

- (c) Given that $x^4 + 3x^2 - 100 = 0$ has a root near $x = 3$, use Newton's method once to find a better approximation, giving your answer in exact form.
- (2M)

- (d) Find the numerical value of the co-efficient of x^0 in the expansion of

$$\left(2x^2 - \frac{1}{2x} \right)^{12} \quad (3M)$$

- (e) Find the sum of the six co-efficients (including the co-efficient of x^0) in the expansion of $(3 - x)^5$
- (2M)

QUESTION 3 (use a SEPARATE writing booklet)

- (a) Using the table of standard integrals, evaluate $\int_0^2 \frac{8dx}{x^2 + 4}$ (2M)

- (b) Find in radians all the values of x in the domain $0 \leq x \leq \pi$ which satisfy the equation

$$\sin 2x - \sin x = 0 \quad (3M)$$

- (c) If α , β and γ are the roots of the cubic equation

$$2x^3 - 8x^2 + x + 12 = 0,$$

write down the value of

- (i) $\alpha + \beta + \gamma$
(ii) $\alpha\beta + \beta\gamma + \gamma\alpha$
(iii) $\alpha^2 + \beta^2 + \gamma^2$ (3M)

- (d)

(i) Show that $\frac{d}{dx}(y^2) = 2y \frac{dy}{dx}$

(ii) Deduce from (i) that if $x^2 + 4y^2 = 5$ then $\frac{dy}{dx} = \frac{-x}{4y}$

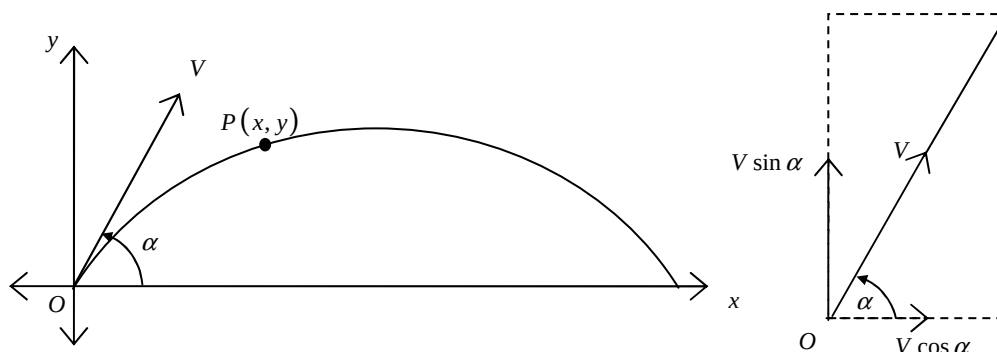
- (iii) A particle moves at a constant speed of $k \text{ ms}^{-1}$ on the circumference of the curve

$$x^2 + 4y^2 = 5$$

Find $\frac{dy}{dt}$ at the point P where $x = 2$, $y = -\frac{1}{2}$ and $\frac{dx}{dt} = 2 \text{ ms}^{-1}$

- (iv) What is the exact value of k , the constant speed? (4M)

QUESTION 4 (use a SEPARATE writing booklet)



(The above diagrams may help in your presentation)

A particle is projected with a velocity V from a point O at an angle of α to the horizontal, in the x, y plane. In the usual notation using the calculus or otherwise, prove that the horizontal distance x and the vertical distance y , travelled by the particle at time t are given by:

(i) $x = (V \cos \alpha)t$ (2M)

(ii) $y = (V \sin \alpha)t - \frac{1}{2}gt^2$
where g is the acceleration due to gravity. (3M)

(iii) By combining (i) and (ii) deduce that:

$$y = x \tan \alpha - \frac{gx^2}{2V^2}(1 + \tan^2 \alpha) \quad (3M)$$

(iv) Any of the above results may be used in attempting this problem:

A football is kicked at 15ms^{-1} and just passes over a crossbar 5m high and 15m away.

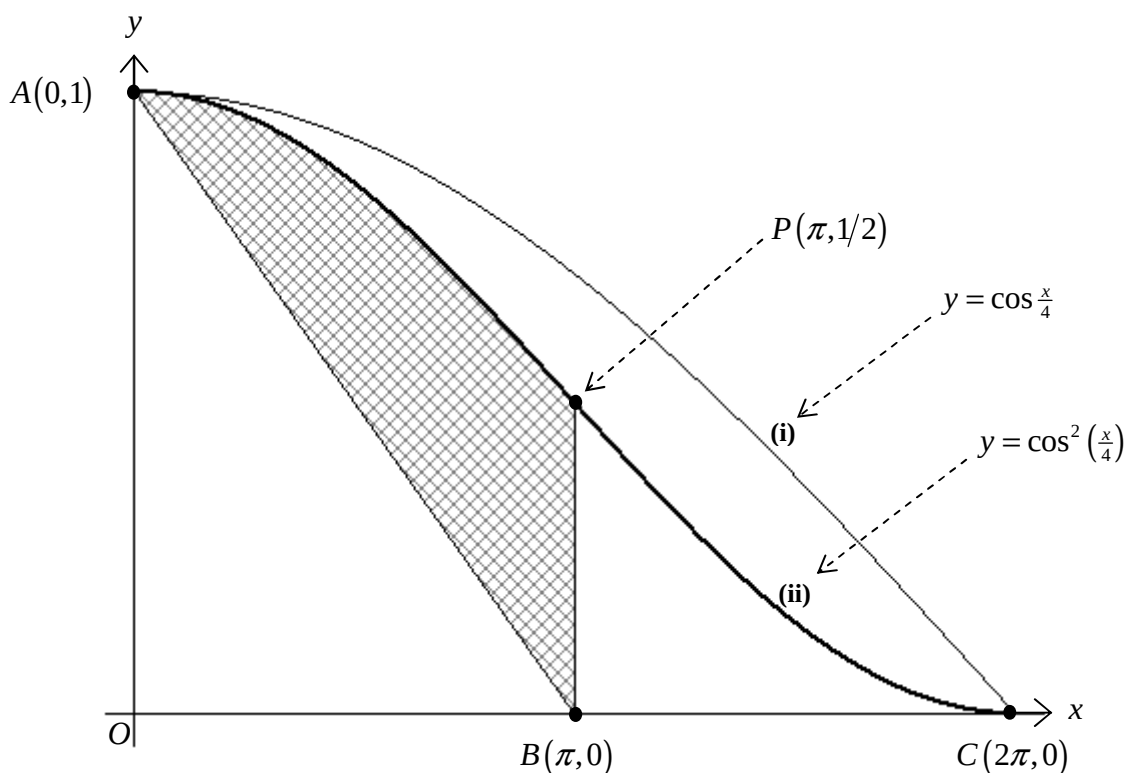
Taking $g = 10\text{ms}^{-2}$ vertically downwards, show that if α is the angle of projection then $\alpha = \frac{\pi}{4}$ or $\alpha = \tan^{-1} 2$

(3M)

(v) If the two values of α (from part iv) are given by A and B (respectively) where $A > B$, write down the exact value of $\tan(A - B)$

(1M)

QUESTION 5 (use a SEPARATE writing booklet)



The above diagram shows two curves

$$y = \cos \frac{x}{4} \quad \text{(i) (thin line)}$$

$$y = \cos^2 \left(\frac{x}{4} \right) \quad \text{(ii) (thick line)}$$

Also, A is the point $(0,1)$, B is the point $(\pi,0)$ and C is the point $(2\pi,0)$

- (i) Prove that the area enclosed by the curve $y = \cos \frac{x}{4}$ and the x and y axes is 4 square units.

(2M)

- (ii) The shaded region $APBA$ is enclosed by $y = \cos^2 \left(\frac{x}{4} \right)$ and the lines AB and $x = \pi$. Prove that the area of this shaded region is 1 square unit.

(3M)

- (iii) Prove that $\cos^4 \left(\frac{x}{4} \right) = \frac{1}{8} \left(3 + 4 \cos \frac{x}{2} + \cos x \right)$

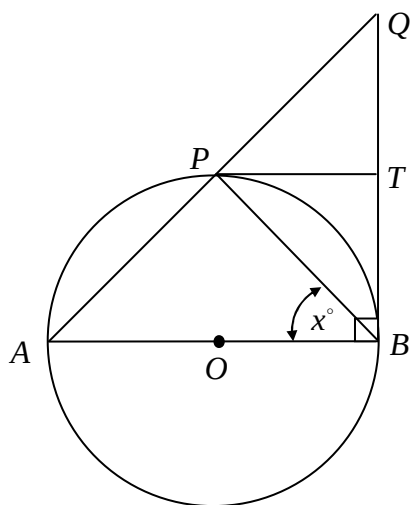
(3M)

- (iv) The shaded region is now rotated through 2π radians about the x -axis. Show, using the result of part (iii) or otherwise that the exact volume of the solid generated is $\pi \left(1 + \frac{\pi}{24} \right)$ cubic units.

(4M)

QUESTION 6 (use a SEPARATE writing booklet)

- (a) If $y = 2007^x$, find an expression for $\frac{dy}{dx}$ in terms of x (2M)



- (b) AB is the diameter of a circle, centre O . P is a point on the circumference. The tangents at B and P meet at T , and AP , BT are produced to meet at Q , and $\hat{ABP} = x^\circ$

- Copy the diagram into your script and explain why $\hat{APB} = 90^\circ$
- Noting without any explanation, that $\hat{OBQ} = 90^\circ$ or otherwise, explain why $\hat{TQP} = x^\circ$
- Explain why $\hat{TPQ} = x^\circ$
- If $x = 30$ show $PQ = 3AP$ (4M)

(c)

- In the usual notation, prove that $\ddot{x} = v \frac{dv}{dx}$ (1M)

- A particle moves in a straight line and its acceleration at any time t is given by $\ddot{x} = v \frac{dv}{dx} = -e^{-2x}$, where x is the displacement and v the velocity at time t . Also, when $x = 0$, $v = 1$.

By starting with the result of part (i) or otherwise, prove that $v = e^{-x}$. (3M)

- It is also known that when $t = 0$, $x = 0$. Deduce from (ii), or otherwise prove, that the displacement x at time t is given by $x = \ln(t+1)$ (2M)

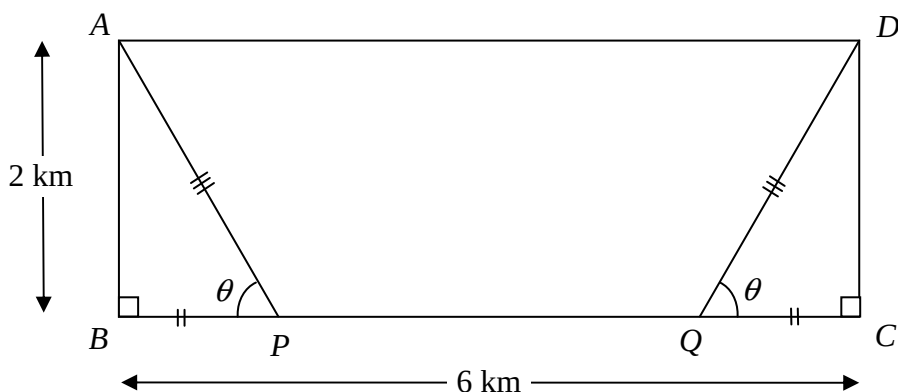
QUESTION 7 (use a SEPARATE writing booklet)

- (a) If $y = 3 \cos^{-1} \left(\frac{x}{2} \right)$, write down (i) the domain of x
(ii) the range of y (2M)

- (b) If $y = f(x) = \frac{kx+l}{mx-k}$ (where k, l, m are constants), prove that $x = f(y)$ (2M)

- (c) In attempting the problem below, you may assume without proof that:

- If $y = \cot \theta$, $\frac{dy}{d\theta} = -\operatorname{cosec}^2 \theta$
- If $y = \operatorname{cosec} \theta$, $\frac{dy}{d\theta} = -\operatorname{cosec} \theta \cot \theta$



The diagram shows a house at A , a school at D and a straight canal BC , where $ABCD$ is a rectangle with $AB = 2\text{ km}$ and $BC = 6\text{ km}$.

During the winter, when the canal freezes over, Danny travels from A to D by walking to a point P on the canal, skating along the canal to a point Q and then walking from Q to D . The points P and Q are chosen so that the angles APB and DQC are both equal to θ , $AP = QD$, $BP = QC$.

- (i) Show that $PQ = (6 - 4 \cot \theta)$ (1M)
- (ii) Given that Danny walks at a constant speed of 4 km h^{-1} , and skates at a constant speed of 8 km h^{-1} show that the time, T minutes, taken for Danny to go from A to D along this route is given by
- $$T = 15(3 + 4 \operatorname{cosec} \theta - 2 \cot \theta) \quad (3\text{M})$$
- (iii) Show clearly and carefully that, as θ varies, Danny's minimum time for the journey is approximately 97 minutes. (4M)

END OF PAPER

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

QUESTION 1

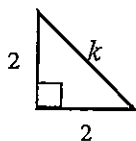
Suggested Solutions	Max Mark	Your Mark	Marker's Comments
(a) (i) $\frac{1}{\sqrt{1-x^2}}$ $x < 1$ (ii) $\frac{1}{\sqrt{49-x^2}}$ $x < 7$ (iii) $\frac{1}{\sqrt{\left(\frac{1}{7}\right)^2 - x^2}}$ OR $\frac{7}{\sqrt{1-49x^2}}$ $x < \left \frac{1}{7}\right$ No penalty for omitting domain of x	2 1		
(b) (i) Here $f(3) = 27 - 27 - 12 + 12 = 0$ (ii) From (i) $f(x) = (x-3)(x^2-4)$ Here $f(x) = (x-3)(x-2)(x+2) = 0$ Hence the required solutions are $x = 3, 2, -2$ *	1 1 1	✓ ✓ ✓	* No mark awarded if soln. not written.
(c) L.H.S. = $\frac{1}{3}[n(n+1)(n+2) + 3(n+1)(n+2)]$ We take out the factors $[(n+1)(n+2)]$ To get L.H.S. = $\frac{1}{3}[(n+1)(n+2)](n+3)$ (Award 2 marks for L.H.S. = R.H.S by direct multiplication) $\frac{1}{3}(n^3 + 6n^2 + 11n + 6)$	1 1	✓ ✓	Some tried Induction. (+) for 1st two steps; if no solution was shown.
(d) $\frac{x-2}{x+4}$ is undefined if $x+4=0$ ie if $x=-4$ Multiply both sides by $3(x+4)^2$ To get $3(x-2)(x+4) \geq (x+4)^2$, ($x \neq -4$) ie $3x^2 + 6x - 24 \geq x^2 + 8x + 16$, ($x \neq -4$) Hence $x^2 - x - 20 \geq 0$, ($x \neq -4$) ie $(x+4)(x-5) \geq 0$, but $x \neq -4$ $\therefore x < -4$ or $x \geq 5$ To gain the 4th mark, the solution needs to mention $x \neq -4$ just the once.	1 1 1 1	✓ ✓ ✓ ✓	If $x \neq -4$ had to be mentioned, otherwise -1.

QUESTION 2

Suggested Solutions	Max Mark	Your Mark	Marker's Comments
(a) (i) $\dot{x} = 10 \cos\left(5t + \frac{\pi}{6}\right)$ ✓ (ii) $\ddot{x} = -50 \sin\left(5t + \frac{\pi}{6}\right)$ ✓ (iii) $\ddot{x} = -5^2 \left(2 \sin\left(5t + \frac{\pi}{6}\right)\right) = -5^2 x \quad \otimes$ At least 2 explanations are acceptable Eg (α) $x = 2 \sin\left(5t + \frac{\pi}{6}\right) = 2 \cos\left(5t - \frac{\pi}{3}\right)$, defines S.H.M (β) Equation $\otimes \quad \ddot{x} = -n^2 x \quad (n=5)$ defines S.H.M } ✓	1		
(b) Here $\frac{dy}{dx} = \frac{dy}{dt} \cdot \frac{dt}{dx} = 4t \times \frac{1}{2} = 2t$ ✓ OR $\frac{dy}{dx} = 2$ At P, $t=1 \therefore$ Equation is $y = 2x - 2$ OR $2x - y - 2 = 0$ ✓	1		most students used $\ddot{x} = -n^2 x$
(c) Here $f(x) = x^4 + 3x^2 - 100 \therefore f(3) = 8$ $f'(x) = 4x^3 + 6x \quad f'(3) = 126$ } both correct ✓	1		
In the usual notation $x_1 = x_0 - \frac{f(x)}{f'(x)} = 3 - \frac{8}{126}$ $= 2 \frac{59}{63} \quad (= \frac{185}{63})$ ✓	1		
(d) (i) Here $T_{R+1} = \binom{12}{R} \left(\frac{-x^{-1}}{2}\right)^R \cdot (2x^2)^{12-R}$ ✓ Power of x in $T_{R+1} = -R + 24 - 2R$ $= 24 - 3R = 0$ when $R = 8$ ✓ So $T_9 = \binom{12}{8} \left(\frac{1}{2^8}\right) 2^4 = \frac{12 \cdot 11 \cdot 10 \cdot 9}{4 \cdot 3 \cdot 2 \cdot 1} \times \frac{1}{16}$ $= \frac{495}{16} \quad (= 30.9353 = 30 \frac{15}{16})$ ✓	1		
(ii) To sum the 6 co-efficients, set $x = 1$ Required sum $(3 - 1)^5 = 2^5 = 32$ OR ✓ $3^5 + \binom{5}{1} 3^4 (-1) + \binom{5}{2} 3^3 (-1)^2 + \binom{5}{3} 3^2 (-1)^3 + \binom{5}{4} 3 (-1)^4 + (-1)^5$ ✓	1		
	1		

QUESTION 3

Suggested Solutions	Max Mark	Your Mark	Marker's Comments
(a) $I = \left[4 \tan^{-1} \left(\frac{x}{2} \right) \right]_0^2$ $= 4 \left[\tan^{-1} 1 - \tan^{-1} 0 \right] = \frac{4\pi}{4} = \pi$	1	✓	
	1	π ✓	
(b) $2 \sin x \cos x - \sin x = 0$ $\therefore \sin x (2 \cos x - 1) = 0 \rightarrow \sin x = 0$ or $\cos x = \frac{1}{2}$ So $x = 0, \pi$ or 2π or $x = \frac{\pi}{3}$ or $\frac{5\pi}{3}$ But $0 \leq x \leq \pi$ $x = 0, \frac{\pi}{3}, \pi$ (Penalty of 1 if degrees are used instead of radians)	1	✓	
	1	✓	
	1	0, $\frac{\pi}{3}$, π ✓	Any answers other than these three did not get a mark here.
(c) (i) 4 (ii) $\frac{1}{2}$	1	4 ✓	
	1	$\frac{1}{2}$ ✓	
(iii) $\alpha^2 + \beta^2 + \gamma^2 = (\alpha + \beta + \gamma)^2 - 2(\alpha\beta + \beta\gamma + \alpha\gamma) = 15$	1	15 ✓	
(d) (i) Use product rule $\frac{d}{dx}(y \cdot y) = yy' + y' y = 2yy'$ OR note $\frac{d}{dx}(y^2) = \frac{d}{dy}(y^2) \cdot \frac{dy}{dx} = 2yy'$ (ii) Differentiate both sides with respect to x To get $2x + 8y \cdot \frac{dy}{dx} = 0 \therefore \frac{dy}{dx} = \frac{-x}{4y}$ (iii) $\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt} = \frac{-x}{4y} (2) = 2ms^{-1}$ (iv) Note: we focus on <u>speed</u> (and not velocity) At P, speed along x axis = speed along the y axis $= 2ms^{-1} \therefore$ Resulting speed $= \sqrt{2^2 + 2^2}$ $k = 2\sqrt{2}$	1	$2y \frac{dy}{dx}$ ✓	Some tried to reproduce the Question - No marks. All other methods were accepted.
	1	$\frac{-x}{4y}$ ✓	
	1	2 ✓	
	1	$2\sqrt{2}$ ✓	



Suggested Solutions

Your Mark

Marker's Comments

The initial velocity in a horizontal direction is $V \cos \alpha$ and since no forces are acting in the horizontal direction, this velocity will remain uniform. $\ddot{x} = 0$

That is $\frac{dx}{dt} = V \cos \alpha$ ✓

Thus $x = (V \cos \alpha)t + c$

But $x = 0$ when $t = 0$, thus $c = 0$

Hence $x = (V \cos \alpha)t \dots(i)$ ✓

The initial velocity in a vertical direction is $V \sin \alpha$. If we take the upwards direction as positive the particle experiences a uniform acceleration of $-g$ due to gravity.

Thus $\frac{d^2 y}{dt^2} = -g$ ✓

Hence $\frac{dy}{dt} = -gt + c$

But $\frac{dy}{dt} = V \sin \alpha$ when $t = 0$, thus $c = V \sin \alpha$

Thus $\frac{dy}{dt} = V \sin \alpha - gt$ ✓

Hence $y = (V \sin \alpha)t - \frac{1}{2}gt^2 + k$

But $y = 0$ when $t = 0$, thus $k = 0$

Giving $y = (V \sin \alpha)t - \frac{1}{2}gt^2 \dots(ii)$ ✓

From (i) $t = \frac{x}{V \cos \alpha}$ ✓

Substituting in (ii) $y = \frac{(V \sin \alpha)x}{V \cos \alpha} - \frac{1}{2}g\left(\frac{x}{V \cos \alpha}\right)^2$ ✓

$$\text{i.e. } y = x \tan \alpha - \frac{gx^2}{2V^2} \sec^2 \alpha = x \tan \alpha - \frac{gx^2}{2V^2} (1 + \tan^2 \alpha) \dots \text{(iii)}$$

(iv) Using equation (iii) $5 = 15 \tan \alpha - 5 - 5 \tan^2 \alpha$,

⑦ $5 = 15t - 5 - 5t^2$ where $t = \tan \alpha$

$$\therefore 5t^2 - 15t + 10 = 0$$

$$t^2 - 3t + 2 = 0 \quad \checkmark$$

$$(t-2)(t-1)=0$$

Hence $t=1$ and $t=2$

$$\therefore \alpha = \frac{\pi}{4} \text{ \& } \alpha = \tan^{-1} 2 \quad \checkmark$$

$$(v) \tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B} = \frac{2 - 1}{1 + 2} = \frac{1}{3}$$

You must show the initial conditions to work out the constants.

$$\sec^2 \alpha = 1 + \tan^2 \alpha$$

1st mark :
correct substitution

2nd mark :
correct equation

N.B question
A > B $\therefore$ answer = $\frac{1}{3}$

MATHEMATICS EXTENSION 1
Solutions, Marking Scheme & Comments

QUESTION 5

Suggested Solutions	Max Mark	Your Mark	Marker's Comments
(i) From diagram, Area = $\int_0^{2\pi} \cos \frac{x}{4} dx$	1	✓	
i.e. $A = \left[4 \sin \frac{x}{4} \right]_0^{2\pi} = 4 \sin \frac{\pi}{2} - 0 = 4$	1	✓	
(ii) Shaded Area = $\int_0^{\pi} \cos^2 \left(\frac{x}{4} \right) dx - \Delta BOA$ $= \frac{1}{2} \int_0^{\pi} \left(\cos \frac{x}{2} + 1 \right) dx - \frac{1}{2} (\pi \times 1)$ $= \frac{1}{2} \left[2 \sin \left(\frac{x}{2} \right) + x \right]_0^{\pi} - \frac{\pi}{2}$ $= \left[\sin \frac{\pi}{2} + \frac{\pi}{2} - 0 - 0 \right] - \frac{\pi}{2} = 1$	1	✓	
(iii) Since $\cos^2 \left(\frac{x}{4} \right) = \frac{1}{2} \left[1 + \cos \left(\frac{x}{2} \right) \right]$ Therefore $\cos^4 \frac{x}{4} = \frac{1}{4} \left[1 + \cos \left(\frac{x}{2} \right) \right]^2$ $\therefore \cos^4 \frac{x}{4} = \frac{1}{4} \left[\left(1 + 2 \cos \frac{x}{2} + \cos^2 \left(\frac{x}{2} \right) \right) \right]$ $= \frac{1}{4} \left[1 + 2 \cos \left(\frac{x}{2} \right) + \frac{1}{2} (1 + \cos x) \right]$ $= \frac{1}{8} \left(3 + 4 \cos \frac{x}{2} + \cos x \right)$	1 1 1	✓ ✓ ✓	$\cos^2 A = \cos 2A + 1$ $\frac{1}{2} \left[2 \sin \left(\frac{\pi}{2} \right) + \pi \right]_0^{\pi} - \frac{\pi}{2}$ $= \frac{1}{2} \left[2 \times \sin \frac{\pi}{2} + \pi - 0 \right] - \frac{\pi}{2}$ $= \frac{1}{2} \left[2 + \pi - 0 \right] - \frac{\pi}{2}$ $= \frac{2 + \pi}{2} - \frac{\pi}{2} = 1$
(iv) Required Volume = $\pi \int_0^{\pi} \cos^4 \left(\frac{x}{4} \right) dx - v_c$ where v_c = volume of a cone with radius unity and height π . ie $v_c = \frac{1}{3} \cdot \pi \cdot 1^2 \cdot \pi = \frac{\pi^2}{3}$ $\therefore$ Required Volume = $\frac{\pi}{8} \int_0^{\pi} \left(3 + 4 \cos \frac{x}{2} + \cos x \right) dx - \frac{\pi^2}{3}$ from (iii) $= \frac{\pi}{8} \left[3x + 8 \sin \frac{x}{2} + \sin x \right]_0^{\pi} - \frac{\pi^2}{3}$ $= \frac{\pi}{8} [3\pi + 8 + 0] - \frac{\pi}{8} [0 + 0 + 0] - \frac{\pi^2}{3}$ $= \frac{8\pi}{8} + \pi^2 \left[\frac{3}{8} - \frac{1}{3} \right] = \pi + \frac{\pi^2}{24} = \pi \left[1 + \frac{\pi}{24} \right]$	1 1 1 1 1	✓ ✓ ✓ ✓ ✓	$\rightarrow \frac{3\pi^2}{8} + 1 - \frac{\pi^2}{3}$ Few got to the end yet if 4 right steps full mark was awarded.

Suggested Solutions

**Max
Mark**

Your Mark

Marker's Comments

differentiate both sides with respect to x : $\frac{1}{y} \cdot \frac{dy}{dx} = \ln 2007$

now multiply both sides by $y = (2007)^x$ to get $\frac{dy}{dx} = (2007)^x (\ln 2007)$

(b) (i) $\hat{APB} = 90^\circ$ since it is the angle in the semi circle diameter AB

$$(ii) \quad \hat{O\hat{B}Q} = 90^\circ \text{ (given)}$$

$$\therefore \hat{PBQ} = 90^\circ - x^\circ$$

But $\hat{APB} = \hat{PBQ} + \hat{QPB}$

(Ext $\angle$ of Δ = sum of interior opp $\angle$'s)

$$90^\circ = 90^\circ - x^\circ + \hat{PQB}$$

$$\therefore P\hat{Q}B = x^\circ = T\hat{Q}P \text{ (same angle)}$$

(iii) $TP=TB$ (tangents from ext point to a circle are equal)

$$\hat{TPB} = \hat{BTP} = 90^\circ - x^\circ$$

$$\text{So } \hat{TPQ} = [90^\circ - (90^\circ - x^\circ)] = x^\circ$$

$$(iv) \quad \frac{AP}{AQ} = \frac{AB \cos 60^\circ}{\left(\frac{AB}{\cos 60^\circ} \right)} = \left(\frac{1}{2} \times \frac{1}{2} \right) = \frac{1}{4} \quad \therefore PQ = 3AP$$

(c) (i) $\ddot{x} = \frac{dv}{dt} = \frac{dx}{dt} \cdot \frac{dv}{dx} = v \frac{dv}{dx}$

(ii) Using part (i) $v \frac{dv}{dx} = -e^{-2x}$ and rearranging, $\int v dv = -\int e^{-2x} dx$

Thus $\frac{v^2}{2} = \frac{1}{2}e^{-2x} + K$ Set $x = 0$ and $v = 1$

To get $\frac{1}{2} = \frac{1}{2}(1) + K \Rightarrow K = 0$ thus $v^2 = e^{-2x}$ so $v = e^{-x}$

(iii) From part (ii) $\frac{dx}{dt} = e^{-x}$

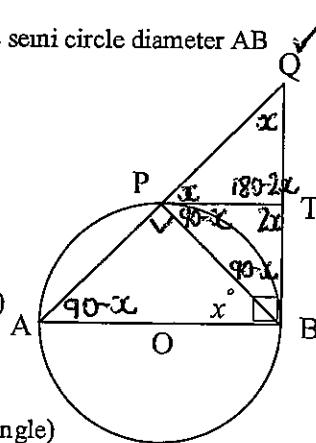
So again rearranging, $\frac{dx}{e^{-x}} = dt$

and hence $\int e^x dx = \int dt \Rightarrow e^x = t + c$

set $t=0$ and $x=0$, (data), $e^0 = 0 + c \quad \therefore c=1$

ie $e^x = t + 1$

Finally take logs of each side to base e, to get $x = \ln|t + 1|$



poorly answered
some Extension 2
students O.K.

Note: You cannot assume $PT \parallel AB$

I have written in the correct values for each angle in terms of the pronumerals.

* Some used $\frac{d}{dx}(\frac{1}{2}v^2) = \ddot{x}$
This was O.K.

Note:
 $\frac{dx}{dt} = e^{-x}$
 to assist in
 correct integration

QUESTION 7

$-2 \leq x \leq 2$

Suggested Solutions

Max
Mark

Your
Mark

Marker's Comments

(a) (i) $x \leq |2|$ (ii) $0 \leq y \leq 3\pi$

1+1

so many

disappointing that
students didn't get
these correct.

(b) Note that $mxy - ky = kx + l \therefore mxy - kx = ky + l$ and so

$x(my - k) = ky + l \Rightarrow x = \frac{ky + l}{my - k} = f(y)$

1

→

attempting to make
the subject

1

→

correct result

(neither x , or $y = \frac{k}{m}$)

(c) (i) $PQ = BC - 2BP$ and in right angled $\triangle ABP$

$\tan \theta = \frac{2}{BP} \therefore \cot \theta = \frac{BP}{2} \therefore BP = 2 \cot \theta$

Hence $PQ = BC - 2(2 \cot \theta) = 6 - 4 \cot \theta$

1

(ii) By similar reasoning as in (i)

$AP = 2 \operatorname{cosec} \theta$ so $AP + QD = 4 \operatorname{cosec} \theta$

Now focus on time in MINUTES (t_m)

t_m for D to travel $AP + QD = 60 \left(\frac{4 \operatorname{cosec} \theta}{4} \right) = 60 \operatorname{cosec} \theta$

t_m for D to travel $PQ = 60 \left(\frac{6 - 4 \cot \theta}{8} \right) = (45 - 30 \cot \theta)$

Let $T = \text{Total } t_m$, so $T = 60 \operatorname{cosec} \theta + 45 - 30 \cot \theta$
 $= 15(3 + 4 \operatorname{cosec} \theta - 2 \cot \theta)$

(iii) $\therefore \frac{dT}{d\theta} = -60 \operatorname{cosec} \theta \cot \theta + 30 \operatorname{cosec}^2 \theta$

$= 30 \operatorname{cosec}^2 \theta (-2 \cos \theta + 1)$ Note $30 \operatorname{cosec}^2 \theta > 0$

So (in discussing) the sign of $\frac{dT}{d\theta}$ we need focus on sign of $(-2 \cos \theta + 1)$

Now for a Max or a Min $\frac{dT}{d\theta} = 0$ so $-2 \cos \theta + 1 = 0$ or $\cos \theta = \frac{1}{2}$ and

$\theta = \frac{\pi}{3}$ (Note θ is acute)

If $\theta = \frac{\pi}{3}$, $T = 15 \left(\frac{8}{\sqrt{3}} + 3 - \frac{2}{\sqrt{3}} \right) = 45 + 30\sqrt{3}$

$\approx 45 + 51.963 \approx 96.963...$

Hence for $\theta = \frac{\pi}{3}$, T is just slightly below 97 minutes. We have now to show this is

a minimum. Note $\frac{dT}{d\theta}$ takes the same sign as $(1 - 2 \cos \theta)$

Now if $0 \leq \theta < \frac{\pi}{3}$, $\cos \theta > \frac{1}{2} \therefore 1 - 2 \cos \theta < 0$

Now if $\frac{\pi}{3} < \theta \leq \frac{\pi}{2}$, $\therefore \cos \theta < \frac{1}{2}$, $\therefore 1 - 2 \cos \theta > 0$

Hence as θ passes through $\frac{\pi}{3}$, $\frac{dT}{d\theta}$ goes from negative through 0 to positive.

Hence at $\theta = \frac{\pi}{3}$ we get the least value of T .

when $\theta = \frac{\pi}{3}$, $T \doteq 97$ minutes

1

→

correct expression
for length of AP or QD.

1

→

correct expression for
time taken (even if in
hours)

1

→

correct conversion from
time in hours to time
in minutes.

1

→

correct expression
for $\frac{dT}{d\theta}$

1

→

* many students didn't
use the given results at
beginning of question.

1

→

correct solution
for $\theta \left(\frac{\pi}{3} \right)$

1

→

testing that
 $\theta = \frac{\pi}{3}$ gives a
minimum time

1

→

substituting $\theta = \frac{\pi}{3}$
into expression for T
and obtaining $T \doteq 97$ min.

QUESTION 7 (c) (iii)

Suggested Solutions	Max Mark	Your Mark	Marker's Comments
2nd way of showing that $\theta = \frac{\pi}{3}$ gives a minimum time for Danny's journey. Note that $\frac{1}{30} \frac{dT}{d\theta} = \operatorname{cosec}^2 \theta (1 - 2 \cos \theta)$ $\therefore \frac{1}{30} \frac{d^2T}{d\theta^2} = (1 - 2 \cos \theta) [-2 \operatorname{cosec}^2 \theta \cot \theta] + \operatorname{cosec}^2 \theta (2 \sin \theta) \dots (L)$ RHS of L $= [-2 \operatorname{cosec}^2 \theta \cot \theta + 4 \operatorname{cosec}^2 \theta \cot \theta \cos \theta] + 2 \operatorname{cosec} \theta$ $= 2 \operatorname{cosec}^2 \theta \cot \theta (2 \cos \theta - 1) + 2 \operatorname{cosec} \theta$ $= 2 \operatorname{cosec}^2 \frac{\pi}{3} \cot \frac{\pi}{3} (2 \cos \frac{\pi}{3} - 1) + 2 \operatorname{cosec} \frac{\pi}{3}, \text{ if } \theta = \frac{\pi}{3}.$ $= 0 + \frac{4}{\sqrt{3}}, \text{ as } 2 \cos \frac{\pi}{3} = 1$ $= \frac{4}{\sqrt{3}}$ $\therefore \frac{1}{30} \cdot \frac{d^2T}{d\theta^2} = \frac{4}{\sqrt{3}}$ $\therefore \frac{d^2T}{d\theta^2} = 30 \times \frac{4}{\sqrt{3}} = 40\sqrt{3} > 0$ $\therefore \theta = \frac{\pi}{3}$ gives a minimum time for Danny's journey. Source of question 7(c) University of London GCE Pure Mathematics 405, January 1988. 371/405/420 Question 14	1 1 1 1		